

Additional text 2: is kinase depletion by staurosporine explained by drug binding?

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Question

10 μ M staurosporine depletes protein kinases from both co-chaperone pull-downs (main report): the 346 kinases detected shift by -0.7 to -1.1 standard deviations, while the other ATP-binding proteins do not move. Some kinases, notably the PKC family, rise instead. If the depletion reflects the drug binding kinases and releasing them from the chaperone machinery, then (i) kinases that staurosporine does not bind should not be depleted, and (ii) tighter binders should be depleted more.

Data

Staurosporine dissociation constants (Kd) from ChEMBL for three kinome-wide competition-binding studies: Davis et al. 2011 (371 kinases), Karaman et al. 2008 (278) and Fabian et al. 2005 (102). Measurements on mutant or variant constructs (89 records) were excluded, and each kinase was given the median log10 Kd across studies. The studies agree closely (Spearman 0.98 Davis vs Karaman on 277 kinases, 0.95 and 0.93 for the comparisons with Fabian), giving 374 kinases with a consensus Kd (median 62 nM; 30 not bound at 10 μ M). 239-255 of the kinases in each GA_33 staurosporine target have a Kd.

Chaperone client strength comes from Taipale et al. 2014 (*Cell* 158:434, “A quantitative chaperone interaction network reveals the architecture of cellular protein homeostasis pathways”; supplementary table NIHMS605110-supplement-02, provided by hand): LUMIER interaction scores of 713 client proteins against 60 chaperones and co-chaperones, including CDC37, HSP90AB1 and HSP90AA1 (scores above 7 are significant). 512 clients match a UniProt accession by gene symbol. The kinome-wide client survey of Taipale et al. 2012 could not be obtained (its supplement is served behind a browser check).

Results

1. Tighter binders are depleted more. Spearman correlation between log10 Kd and the staurosporine response (positive = tighter binding, stronger depletion):

Target	Kinases with Kd	rho	p	Binders, Kd < 1 μ M	Non-binders, Kd $\geq$ 10 μ M	Non-kinases
35 °C	239	+0.16	0.01	-0.21 (195)	+0.04 (18)	+0.00
37 °C	245	+0.14	0.03	-0.25 (201)	-0.04 (18)	+0.02
43 °C	254	+0.28	7e-6	-0.31 (208)	+0.01 (19)	+0.00

Mean log2 fold change, staurosporine vs control, the two co-chaperones averaged (number of kinases).

The kinases staurosporine does not bind behave like non-kinases (p = 0.68, 0.07 and 0.93 for the difference), unlike the binders. But there are only 18-19 of them, and the difference between binders and non-binders reaches

significance only at 43 °C ($p = 0.07, 0.2$ and 0.02). The control points the right way without proving that binding is required.

2. The relation holds within kinase groups, and binding is graded in the cell. Correcting for kinase group (AGC, CAMK, CMGC, STE, TKL, tyrosine kinases, and others), the partial rank correlation of Kd with depletion is $+0.12$ [$+0.00, +0.24$] at 35 °C, $+0.11$ [$-0.01, +0.23$] at 37 °C and **$+0.24$ [$+0.13, +0.34$]** at 43 °C (bootstrap over kinases). At 10 μ M nearly every binder below 1 μ M would be saturated in a test tube, which predicts a step near micromolar; the data instead follow Kd across the nanomolar range. Occupancy computed with the test-tube Kd correlates more weakly with the response ($r = -0.15$ at 43 °C) than occupancy computed with Kd raised 1,000-fold ($r = -0.26$), as competition with cellular ATP would do.

3. The AGC exception. By Kd bin (37 °C), the tightest binders are depleted most, but the relation is broken by the AGC kinases; without them it falls off steadily with affinity:

Kd	< 1 nM	1-10 nM	10-100 nM	100 nM-1 μ M	1-10 μ M	$\geq 10 \mu$ M
All kinases	-0.71 (19)	-0.09 (43)	-0.25 (81)	-0.21 (58)	-0.13 (26)	-0.04 (18)
Without AGC kinases	-0.82 (16)	-0.24 (35)	-0.29 (66)	-0.22 (52)	-0.13 (26)	-0.04 (18)
AGC kinases	-0.12 (3)	+0.54 (8)	-0.07 (15)	-0.10 (6)		

The AGC kinases that rise sit mostly in the 1-10 nM bin:

AGC kinase	Kd (nM)	Response	AGC kinase	Kd (nM)	Response
PRKG1	32	+4.36	PRKCD	1.5	+0.37
PRKCQ	8.5	+2.24	PRKCA	50	+0.33
PRKCH	4.8	+1.85	PRKCE	0.2	+0.16
PKN2	2.5	+1.52	RPS6KB1	1.3	+0.12
PKN1	1.3	+1.24	AKT2	44	+0.00

These kinases bind staurosporine as tightly as any, yet gain co-chaperone association. Whatever staurosporine does to PKC, PKN and PKG, it is not release. Group by group, CAMK kinases fall most (mean -0.41, median Kd 16 nM) and AGC kinases rise (+0.07, median Kd 27 nM) at similar affinities.

4. Kinases most present in the control pull-downs lose the most. Among kinases, the level in the control pull-down predicts depletion (Spearman -0.22 to -0.34 with level and response taken from disjoint replicates, so not shared noise). In a joint rank model of the kinases:

Target	Kd	Level in control pull-down	R2: group	+ Kd	+ level
35 °C	+0.19 [$+0.04, +0.32$]	-0.39 [$-0.51, -0.26$]	3.9 %	5.3 %	18.8 %
37 °C	+0.17 [$+0.02, +0.33$]	-0.37 [$-0.50, -0.25$]	3.2 %	4.4 %	17.5 %
43 °C	+0.27 [$+0.15, +0.40$]	-0.32 [$-0.44, -0.21$]	9.7 %	14.8 %	25.0 %

Standardised rank coefficients with 95 % bootstrap intervals; R2 in sample.

Affinity and pull-down level contribute independently, and the level is the stronger. Across all proteins the level does not predict the staurosporine response (ρ -0.07; 0.00 from disjoint replicates), and the kinase depletion is unchanged after adjusting for it (-0.60, -0.90 and -0.98 sd).

5. Chaperone client strength: underpowered, and the trend runs the other way. Only about 40 GA_33 kinases have Taipale 2014 scores, because that network focused on other client classes. Among them, stronger CDC37 and HSP90 clients lose *less* co-chaperone association under staurosporine, opposite to what release from the HSP90-CDC37 system would predict:

Spearman rho with the response (n)	CDC37	HSP90AB1	HSP90AA1	HSPA8	HSPB1
35 °C (41)	+0.22 (p 0.16)	+0.21 (0.19)	+0.23 (0.15)	-0.02 (0.91)	+0.10 (0.56)
37 °C (40)	+0.25 (0.11)	+0.26 (0.11)	+0.24 (0.14)	+0.09 (0.58)	+0.10 (0.54)
43 °C (42)	+0.29 (0.06)	+0.31 (0.049)	+0.23 (0.16)	+0.18 (0.27)	+0.10 (0.55)

None survives correction for the 15 tests. The kinase groups show the same direction: TKL and tyrosine kinases are the strongest CDC37 clients (median scores 33 and 10) and barely change, while CMGC, CAMK and STE kinases are weak clients (0.4-2.7) and the most depleted. CDC37 client strength is also unrelated to each kinase's level in the control pull-downs (rho +0.09, +0.10, -0.15), so it gives no support to reading that level as chaperone engagement. A joint model with Kd, level and CDC37 score cannot be estimated reliably on 32-34 kinases.

Interpretation

The kinase depletion behaves as drug-driven release from the co-chaperones would: it follows binding affinity, within kinase groups as well as between them, more strongly at 43 °C, along a graded curve that fits ATP competition in the cell better than a simple saturation model (not formally tested); and the few kinases staurosporine does not bind are not depleted, as far as 18 of them can show. Two further facts sharpen the picture. Kinases that are most present in the co-chaperone pull-downs to begin with lose the most, which is what release would predict if that level reflects how strongly each kinase engages the chaperone machinery. And the AGC kinases with lipid- and nucleotide-sensing regulatory domains go the other way despite tight binding, which marks them as mechanistically different.

Critical evaluation

- **Binding data.** Three independent studies agree at Spearman 0.93-0.98, so the Kd values are not the weak point. They are measured on isolated kinase domains in vitro, not in cells.
- **The non-binder control is small.** 18-19 kinases; it is consistent with release but does not establish that binding is necessary.
- **Effect sizes.** Affinity explains a few percent of the variance among kinases (partial rho 0.11-0.24). Most of the kinase-to-kinase variation lies elsewhere, notably in the pull-down level and group.
- **What the pull-down level means.** It mixes how much of a kinase the cell contains with how strongly it engages the co-chaperone. Reading it as chaperone engagement is a hypothesis; the input lysate would separate the two (level in the pull-down relative to level in the lysate).
- **AGC kinases.** Their rise is clear but unexplained. One possibility to test: ATP-site inhibitors are known to change the regulatory state of some AGC kinases (for example by protecting activation-loop phosphorylation), which could alter their chaperone engagement in the opposite direction.
- **Client strength.** The Taipale 2014 scores cover only ~40 of the kinases. They do not support release from the HSP90-CDC37 system specifically (the trend is the opposite, not significant), nor the reading of pull-down level as chaperone engagement. The co-chaperones measured here, DNAJA1 and DNAJB11, act upstream of HSP90, so the two need not move together. The 2012 kinome-wide survey (about 300 kinases) would give a powered test.

Next steps

1. HSP90-CDC37 client strength for the full kinome (Taipale et al. 2012, *Cell* 150:987): the 2014 table covers too few of these kinases to decide.
2. The input lysate: depletion from the pull-down relative to depletion from the lysate.

Methods files

`code/ad02_kinase_affinity.py`, `code/ad02b_taipale.py`; ChEMBL extracts in `data/external/kinase/`; results in `data/external/kinase/ad02_results.json`.